

5(a)	<i>two from</i> <i>ref.to non-self / foreign (antigen) ;</i> <i>one from</i> activation / AW, of lymphocytes ; <i>ref. to specificity ;</i> <i>ref. to immunological memory ;</i>	2
5(b)(i)	<i>three from</i> (B-) lymphocytes respond / immune response, to self antigens (on muscle cell surface) / antibody produced by, B-lymphocytes / plasma cells ; A failure to distinguish between self and non-self antibody binds to receptor ; antibody has complementary shape to receptor ; A idea of specificity cell signalling molecule / neurotransmitter / AW, unable to bind to receptor ; reactions / AW, within cells, not triggered / described ;	3
5(b)(ii)	<i>three from</i> active site of enzyme Y is, not (fully) complementary / partially complementary ; active site, moulds to fit / becomes fully complementary to, cell signalling molecule ; enzyme-substrate complex forms ; A ESC forms <i>ref. to</i> interaction between substrate and R groups of enzyme amino acids ; lowers activation energy ; products leave the active site ; enzyme molecule unchanged / able to be re-used ; AVP ; e.g. detail of how activation energy is lowered	4
5(c)	<i>three from</i> I statements about locating or diagnosis of disease specific / targeted, therapy / treatment ; AW A only affects, one cell type in the body / B lymphocytes bind to / recognition of, receptors / antigens / CD20, on cell surface (of diseased cells) ; A in context of B lymphocytes or diseased cells monoclonal antibodies, are not recognised as foreign / do not trigger an immune response ; A <i>ref. to</i> humanisation of antibodies example of presence of monoclonal antibody stimulating the immune system / AW ; <i>in context of examples given</i> A stimulates phagocytosis / activates B-cells attach, radioactive substance / drug (to treat / kill, B lymphocytes / diseased cells) ; <i>must be in context</i>	3